

New Keynesian (NK) Model

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Introduction

- RBC model assumes perfect competition in goods and input markets
 - Classical tradition: Fluctuations are natural and efficient responses
 - Implication: Fiscal and monetary policies ineffective for macro conditions
- In MIUF model, money is neutral
- New-Keynesian economics embrace imperfect competition
 - Keynesian principles: Fluctuations are the result of market failures
 - Implication: Fiscal and monetary policies can improve macro conditions
- Eventual convergence in approach, less fights about methodology
 - Micro foundations, nominal rigidities, shocks to demand and supply
 - Main references: Woodford (2003), Galí (2008), Blanchard (2025)

Assumptions

- Closed economy (no imports, no exports)
- No capital since fixed in short run (McCallum and Nelson, 1999)
 - Careful: Here I_t refers to gross nominal interest rate (not investment)
- Firms: Monopolistic competition and price stickiness (friction)
 - Similar but not identical goods sold under **imperfect** competition
 - No fixed costs, so $MC = \text{average TC}$
- HHs: Consumption bundle of a continuum ($j \in [0, 1]$) of goods
- Central bank: New agent in charge of monetary policy
 - In model, price stickiness makes monetary policy non-neutral
 - CB follows a Taylor (1993) rule to stabilize inflation and output gap
 - New shock: Monetary policy shock (in Taylor rule)

Optimization

- Producers of differentiated goods optimize in two steps
 - Minimize TC (taking factor price as given) by choosing production factor
 - Maximize profits by setting price of good (due to market power)
- HHs optimize in two steps
 - Find consumption vector ($C_{jt} \forall j$) that maximizes total C_t
 - Choose utility maximizing combination of C_t, H_t, M_t

HHs: Consumption Bundle

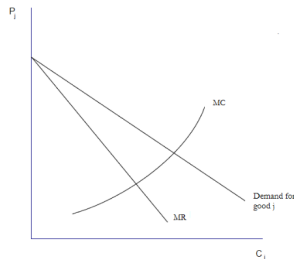
- Common aggregators: Dixit and Stiglitz (1977) and Kimball (1995)
 - Differences discussed by Harding, Lindé and Trabandt (2022)
- Dixit-Stiglitz uses a constant elasticity of substitution (CES) function

$$C_t = \left(\int_0^1 C_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

- CES nests other functions as special cases (ε measures curvature)
 - $\varepsilon \rightarrow \infty$: Goods are perfect substitutes (linear case), no market power
 - $\varepsilon > 1$: Goods are imperfect substitutes, firms are price setters
 - $\varepsilon \rightarrow 1$: Cobb-Douglas case
 - $\varepsilon \rightarrow 0$: Leontief case

Firms: Monopolistic Competition

- Firms produce differentiated goods having some degree of market power
 - Price setters, not price takers
- At optimum,
 - Profit-maximizing firms choose output so that $MR = MC$
 - Since demand curve above MR curve, $price > MC$ (i.e., $markup > 0$)



Firms: Price Stickiness

- Prices do not adjust immediately to changes in demand or supply
 - Rationale: Costly to adjust prices (e.g., menu costs)
- Price stickiness derived from behavior of optimizing agents
 - Rotemberg (1982) pricing: Firms face quadratic costs to change prices
 - Small price changes are less costly than large ones
 - Calvo (1983) pricing: Staggered price contracts
 - Every period, firms keep prices fixed ($p_{jt} = p_{jt-1}$) with probability θ
 - Price-setting is forward looking (may not be able to adjust in the future)
- Calvo pricing was key to derive NK Phillips curve (Roberts, 1995)
- With **nominal rigidities**, price response spreads out over time

Monetary Policy: Non-Neutrality

- Prices do not adjust immediately due to nominal rigidities
- **Implications:** Changes in short-term nominal interest rate
 - Not matched to changes in expected inflation
 - Lead to variations in real interest rate
 - Affect real quantities through effect on real interest rate
- **Main conclusion:** Non-neutrality of monetary policy in short run
- But all prices (and wages) adjust in long run
 - Economy reverts back to its natural equilibrium

Key Variables and Parameters

- Y_t, C_t : Aggregate output or consumption bundle at time t
- Y_{jt}, C_{jt} : Output or consumption by agent j at time t , $\forall j \in [0, 1]$
- B_{t-1} : Zero-coupon bonds bought at $t - 1$ (payoff of 1 at t)
- p_t : Aggregate price level at time t
- p_{jt} : Price of good j at time t
- μ_{jt} : Marginal cost for good j at time t
- s_t : Monetary policy shock
- ε : Parameter for elasticity of substitution between goods
- θ : Parameter for price stickiness (highest when $\theta = 1$)
- ϕ_Y, ϕ_π : Sensitivities of interest rate to output and inflation in Taylor rule

Traditional NK Model

- Dynamic IS curve:

$$\tilde{Y}_t = \mathbb{E}_t \tilde{Y}_{t+1} - \frac{1}{\gamma} (\tilde{i}_t - \mathbb{E}_t \tilde{\pi}_{t+1}) + \mathbb{E}_t \Delta \tilde{A}_{t+1}$$

- NKPC:

$$\tilde{\pi}_t = \beta \mathbb{E}_t \tilde{\pi}_{t+1} + \kappa \tilde{Y}_t + \tilde{u}_t$$

- Taylor rule:

$$\tilde{i}_t = \bar{i} + \phi_\pi \tilde{\pi}_t + \phi_Y \tilde{Y}_t + \tilde{s}_t$$

Dynamic IS (DIS) Curve

$$\tilde{Y}_t = \mathbb{E}_t \tilde{Y}_{t+1} - \frac{1}{\gamma} (\tilde{i}_t - \mathbb{E}_t \tilde{\pi}_{t+1}) + \mathbb{E}_t \Delta \tilde{A}_{t+1}$$

- AD equation:
 - Negative relationship between real interest rate and real activity
- DIS determines output gap given a path for real interest rate
- NK mechanism:
 - If monetary policy influences real rate, it can affect AD

NK Phillips Curve (NKPC)

$$\tilde{\pi}_t = \beta \mathbb{E}_t \tilde{\pi}_{t+1} + \kappa \tilde{Y}_t + \tilde{u}_t$$

- AS equation:
 - Positive relationship between inflation dynamics and real activity
 - Inflation increases when output is larger than its natural level
 - Key role for inflation expectations
- NKPC determines inflation given a path for output gap
- Optimally derived from firms with market power and sticky prices
 - Price stickiness governs slope of NKPC (Dynare)
 - When $\theta \rightarrow 0$ (flexible prices), $\kappa \rightarrow \infty$ (monetary policy is neutral)

Taylor Rule

$$\tilde{i}_t = \bar{i} + \phi_\pi \tilde{\pi}_t + \phi_Y \tilde{Y}_t + \tilde{s}_t$$

- It closes the model and determines its behavior
- How monetary policy reacts to deviations in inflation and output gap
- A role for central banks
 - A contractionary shock lowers both inflation and output gap
- Single mandate central bank: $\phi_Y = 0$
 - Blanchard-Kahn condition (Dynare): $\kappa(\phi_\pi - 1) > 0 \implies \phi_\pi > 1$
 - **Taylor principle:** $\phi_\pi > 1$ (CB reacts to inflation proportionally more)

Effects of (Positive) Shocks

- Technology shock:
 - Persistent decline in output gap (output increases less than natural level)
 - More productive firms, so real marginal cost and inflation fall
 - Central bank partly accommodates shock by lowering rates
 - Economy moves along DIS curve (AS curve shifts): output and inflation
- Cost push shock:
 - Inflation goes up, CB responds, real activity contracts
 - Fall in output gap due to output (natural level unaffected by nominal)
- Monetary policy shock:
 - Inflation (expectations) and output (gap) decline, real rate rises
 - Larger effect on output as degree of price stickiness increases (Dynare)